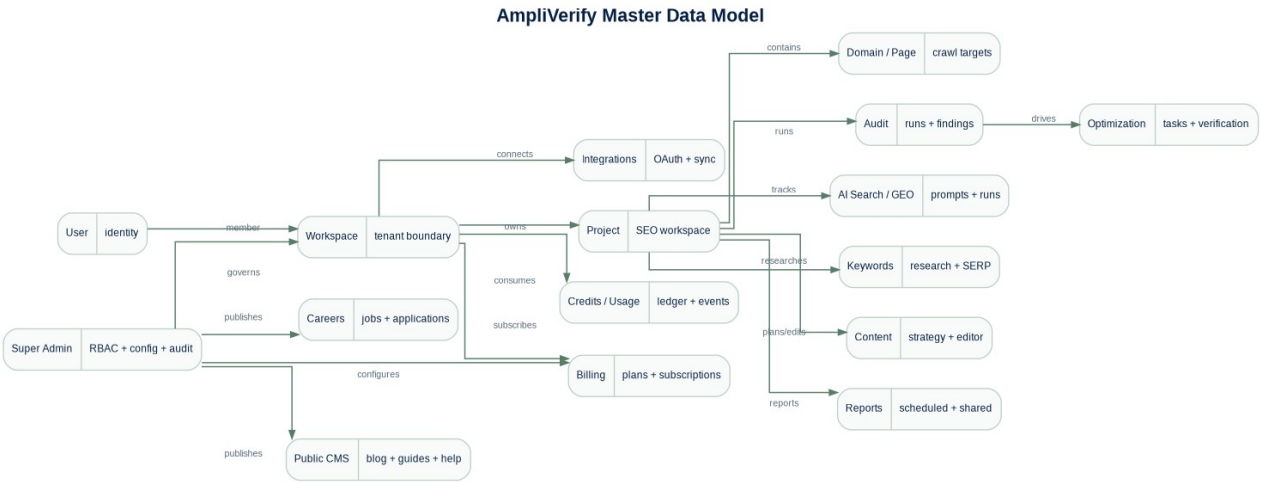


AMPLIVERIFY

DATABASE ARCHITECTURE & IMPLEMENTATION GUIDE

PostgreSQL + Prisma | Multi-tenant SaaS | Public Frontend + User App + Super Admin



Single source of truth for schema, relationships, constraints, security boundaries, indexes, retention, migrations and implementation order.

1. Document Map

Section	What the developer gets
2	Architecture decisions and conventions
3	Master ERD and tenant boundary
4	Identity, workspace and RBAC
5	Projects, domains and pages
6	SEO audit and optimization
7	SEO editor and content strategy
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16	Super Admin, security and audit
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19	Retention, deletion and privacy
20	Migration and implementation order
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23	Proposed Prisma schema core

SOURCE OF TRUTH

The table catalog is the authoritative logical design. Prisma model names may be adjusted to match the final repository conventions, but relationships, tenant scoping and integrity rules should remain intact.

2. Architecture Decisions

Production database direction for the current AmpliVerify product scope.

Decision	Approved direction
Primary database	PostgreSQL 16+
ORM	Prisma
Cache / queues	Redis - not a source of truth
Tenant boundary	workspace_id on tenant-owned records
Project boundary	project_id below workspace where applicable
Authentication	Better Auth / secure cookie sessions; auth adapter tables can remain provider-controlled
Identifiers	UUID or CUID2 consistently; examples use UUID-like IDs
Timestamps	UTC timestamps; created_at and updated_at on mutable records
Soft deletion	Only where restoration/audit value exists; use deleted_at
Financial / security history	Append-oriented; never silently overwrite
External providers	Store provider IDs and references, not provider secrets in plaintext
Large generated artifacts	Object storage; database stores metadata and storage keys
AI responses	Store normalized result metadata and object-storage references for large raw payloads

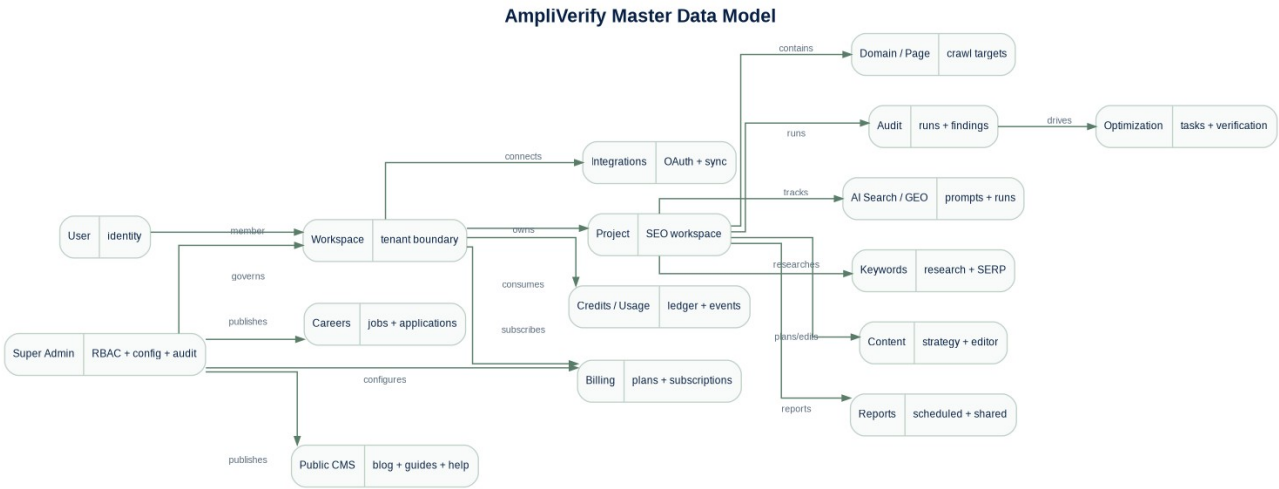
SECURITY

Do not use the frontend as a security boundary. Every protected API query must enforce user identity, workspace membership, permission scope and target-resource ownership server-side.

Naming conventions

- snake_case table and column names in PostgreSQL; Prisma may map PascalCase models with @@map.
- Foreign keys use <entity>_id.
- Boolean names use is_/has_ only when it improves clarity; otherwise use explicit state enums.
- Provider references use provider*_id and remain unique within provider scope.
- Use status enums for workflow state; avoid overloaded booleans for multi-step processes.

3. Master ERD and Tenant Boundary



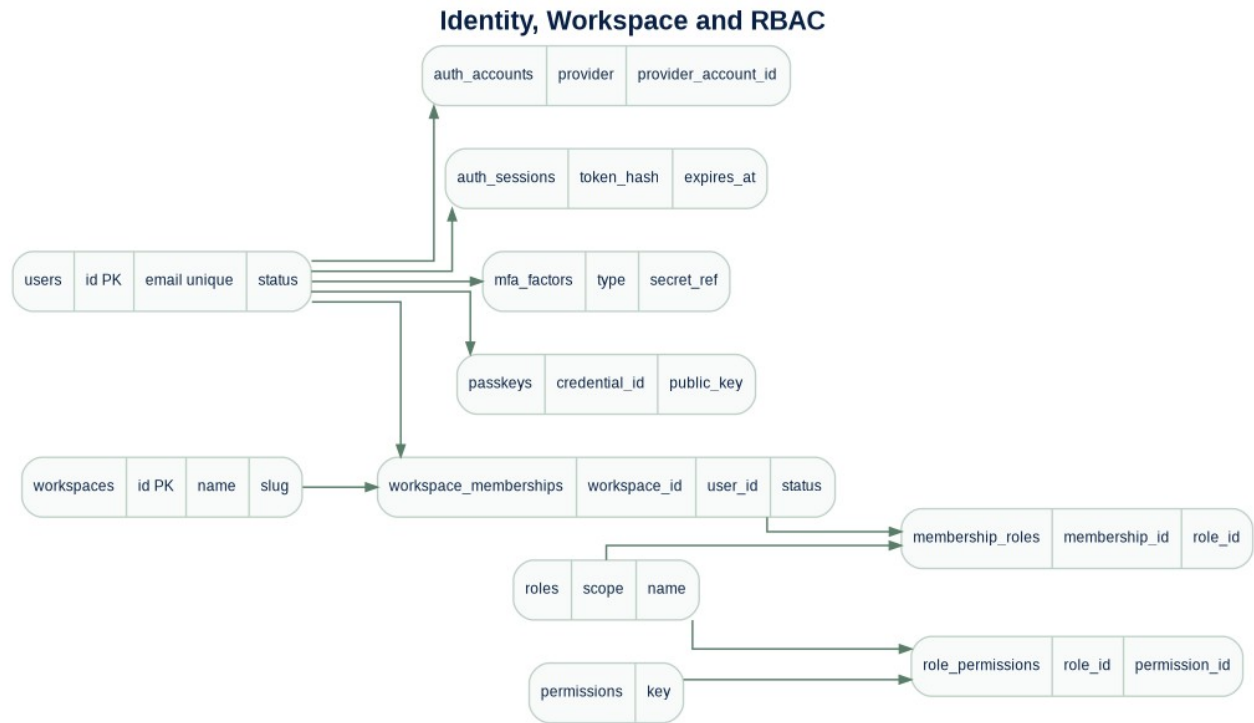
TENANT RULE

Tenant isolation rule: resolve workspace membership first, then constrain every downstream query by workspace_id or by a parent record already constrained to that workspace.

Layer	Primary ownership
Account	users
Tenant	workspaces + workspace_memberships
Product work	projects
Website targets	domains + pages
Analysis	audit_runs / keyword * / geo *
Commercial	subscriptions + plan_entitlements + credit_ledger
Public content	CMS tables
Governance	roles / permissions / audit_logs / feature_flags

4. Identity, Workspace and RBAC

Authentication is identity; authorization is workspace-scoped RBAC enforced by APIs.



AUTH

Let Better Auth own its adapter-specific tables if required. Mirror only application fields you actually need; do not fork authentication internals unnecessarily.

RBAC

System roles may be global templates; custom roles belong to a workspace. Permission checks must resolve membership + role + permission server-side.

Table	Purpose	Key fields
users	Canonical application identity.	id, email, display_name, status, created_at, updated_at...
auth_accounts	OAuth/provider account mapping.	id, user_id, provider, provider_account_id, created_at
auth_sessions	Server-side session metadata.	id, user_id, token_hash, expires_at, ip_hash, user_agent...
mfa_factors	MFA enrollment metadata.	id, user_id, type, secret_ref, verified_at
passkeys	WebAuthn/passkey credentials.	id, user_id, credential_id, public_key, counter, transports
workspaces	Tenant root.	id, name, slug, status, created_by, created_at
workspace_memberships	User-to-workspace membership.	id, workspace_id, user_id, status, joined_at
roles	System or custom role.	id, workspace_id, name, scope, is_system
permissions	Permission registry.	id, key, description
role_permissions	Role permission junction.	role_id, permission_id
membership_roles	Membership role junction.	membership_id, role_id

Integrity and query rules

users: email unique; deleted users retained only per policy. Index: email; status.

auth_accounts: unique(provider, provider_account_id). Index: user_id.

auth_sessions: token_hash unique; never store raw session token. Index: user_id, expires_at.

mfa_factors: one active factor per type unless product supports multiples. Index: user_id.

passkeys: credential_id unique. Index: user_id.

workspaces: slug unique. Index: status.

workspace_memberships: unique(workspace_id,user_id). Index: workspace_id; user_id; status.

roles: unique workspace/name for custom roles. Index: workspace_id; is_system.

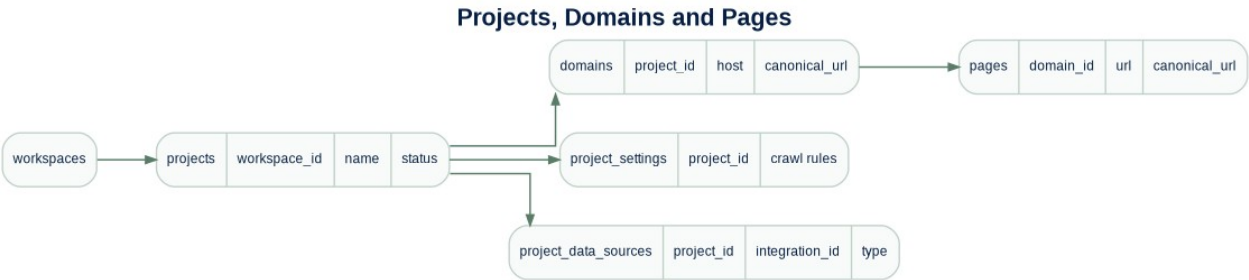
permissions: key unique. Index: key.

role_permissions: composite PK(role_id,permission_id). Index: permission_id.

membership_roles: composite PK. Index: role_id.

5. Projects, Domains and Pages

All SEO work hangs below a workspace-owned project.



URL RULE

Normalize host and URL forms before uniqueness checks. Preserve the fetched URL separately when canonicalization matters.

OWNERSHIP

Do not trust project_id from the client alone. Resolve project -> workspace and confirm the actor has access to that workspace.

Table	Purpose	Key fields
projects	Top-level SEO project within a workspace	id, workspace_id, name, status, created_by, created_at...
domains	Domain/site attached to a project	id, project_id, host, canonical_url, verification_status, created_at
pages	Known page URL	id, domain_id, url, canonical_url, http_status, last_seen_at...
project_settings	Project defaults and crawl policy	project_id, locale, timezone, crawl_config, seo_defaults, updated_at
project_data_sources	Project-to-integration binding	id, project_id, workspace_integration_id, source_type, status

Integrity and query rules

projects: workspace-scoped name may repeat unless product forbids. Index: workspace_id,status.

domains: unique(project_id,host). Index: project_id,host.

pages: unique(domain_id,url). Index: domain_id,last_seen_at.

project_settings: one row per project. Index: project_id.

project_data_sources: unique(project_id,workspace_integration_id,source_type). Index: project_id,status.

6. SEO Audit and Optimization

Audit execution is historical; findings are normalized; optimization converts findings into work.



HISTORY Never overwrite old audit runs. New audits create new run/page/finding records so progress can be compared.

FINGERPRINT Use a stable finding fingerprint per page + rule + context to recognize recurring issues across runs.

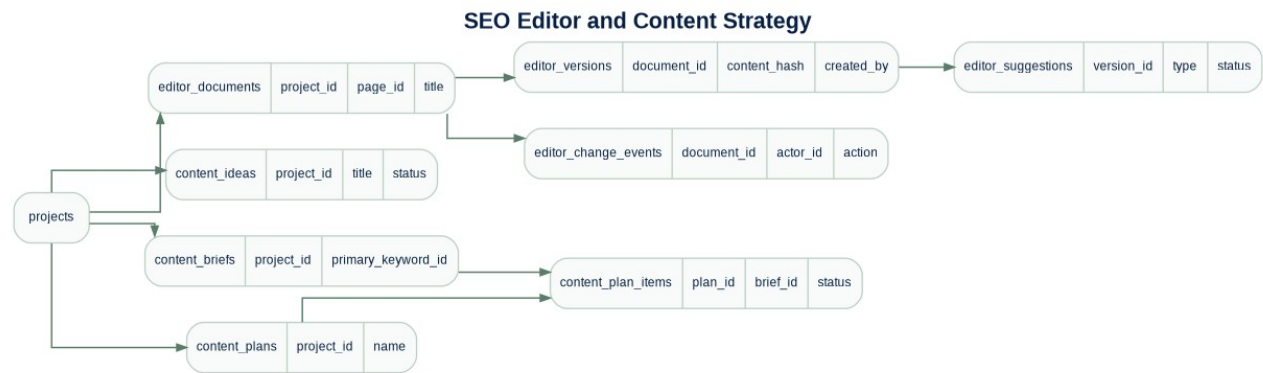
Table	Purpose	Key fields
audit_runs	One execution of an SEO audit.	id, project_id, domain_id, status, trigger, started_at...
audit_pages	Per-page result within a run.	id, audit_run_id, page_id, score, metrics_json
audit_findings	Normalized issue instance.	id, audit_page_id, rule_key, severity, status, fingerprint...
audit_recommendations	Action guidance for finding.	id, finding_id, priority, title, guidance, source
finding_events	Lifecycle history for a finding.	id, finding_id, actor_user_id, event_type, payload, created_at
optimization_tasks	Actionable work item.	id, project_id, finding_id, assigned_to, status, priority...
verification_runs	Verification after implementation.	id, task_id, audit_run_id, status, result, evidence_json...

Integrity and query rules

- audit_runs:** immutable execution identity. Index: project_id,status,started_at desc.
- audit_pages:** unique(audit_run_id,page_id). Index: audit_run_id,page_id.
- audit_findings:** unique(audit_page_id,fingerprint). Index: audit_page_id,severity,status,rule_key.
- audit_recommendations:** one or many recommendations per finding. Index: finding_id,priority.
- finding_events:** append-only. Index: finding_id,created_at.
- optimization_tasks:** tenant-bound through project. Index: project_id,status,assigned_to.
- verification_runs:** append-only verification history. Index: task_id,created_at desc.

7. SEO Editor and Content Strategy

Editor versions are immutable snapshots; strategy objects remain project-scoped.



EDITOR

Store large HTML/document bodies in object storage when size warrants it; persist content hashes and metadata in PostgreSQL.

AI

AI suggestions are advisory records with accepted/rejected state. Do not silently mutate user content.

Table	Purpose	Key fields
editor_documents	Editable SEO document/page workspace.	id, project_id, page_id, title, status, created_by
editor_versions	Immutable document snapshots.	id, document_id, version_no, content_ref, content_hash, created_by...
editor_suggestions	AI/rule suggestions against version.	id, version_id, type, status, suggestion_json, model_ref
editor_change_events	Audit trail for editor actions.	id, document_id, actor_user_id, action, metadata, created_at
content_ideas	Content opportunity/idea.	id, project_id, title, status, source, metadata
content_briefs	Structured brief.	id, project_id, title, primary_keyword_id, brief_json, status
content_plans	Editorial/content plan.	id, project_id, name, status
content_plan_items	Plan item.	id, plan_id, brief_id, title, status, target_date

Integrity and query rules

editor_documents: one active document per page optional, not mandatory. Index: project_id,page_id,status.

editor_versions: unique(document_id,version_no). Index: document_id,version_no desc.

editor_suggestions: never overwrite accepted/rejected history. Index: version_id,status,type.

editor_change_events: append-only. Index: document_id,created_at.

content_ideas: project scoped. Index: project_id,status.

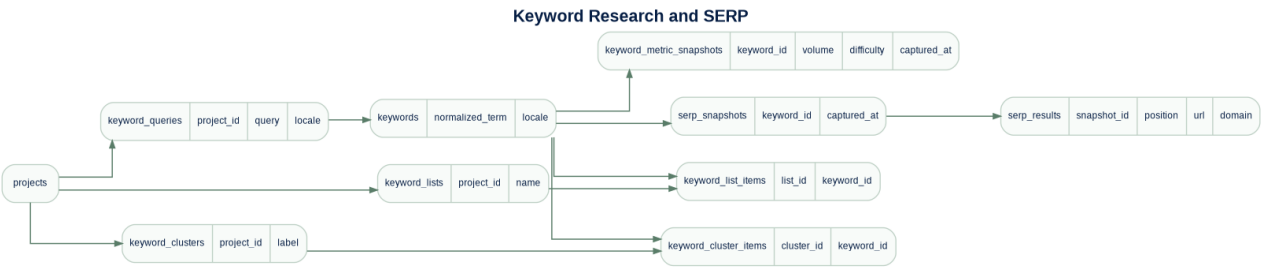
content_briefs: project scoped. Index: project_id,status.

content_plans: project scoped. Index: project_id,status.

content_plan_items: plan scoped. Index: plan_id,status,target_date.

8. Keyword Research and SERP

Normalize reusable keywords, preserve time-series metrics and SERP snapshots.



TIME SERIES

Search volume, difficulty, CPC and SERP positions change. Store snapshots with `captured_at` instead of replacing prior values.

COST

Provider requests should create `usage_events/provider_usage` so the product can meter credits and understand API cost.

Table	Purpose	Key fields
keyword_queries	Research request metadata.	id, project_id, query, locale, country_code, created_by...
keywords	Normalized keyword dictionary.	id, normalized_term, locale, country_code
keyword_metric_snapshots	Time-series keyword metrics.	id, keyword_id, provider, search_volume, difficulty, cpc...
keyword_lists	Saved list.	id, project_id, name, created_by
keyword_list_items	List membership.	list_id, keyword_id, added_at
keyword_clusters	Keyword cluster.	id, project_id, label, method, created_at
keyword_cluster_items	Cluster membership.	cluster_id, keyword_id, score
serp_snapshots	SERP observation.	id, keyword_id, provider, locale, captured_at
serp_results	Ranked result row.	id, snapshot_id, position, url, domain, title...

Integrity and query rules

keyword_queries: request identity immutable. Index: `project_id,created_at`.

keywords: `unique(normalized_term,locale,country_code)`. Index: `normalized_term`.

keyword_metric_snapshots: append-only snapshots. Index: `keyword_id,captured_at desc`.

keyword_lists: `unique(project_id,name)` optional. Index: `project_id`.

keyword_list_items: composite PK. Index: `keyword_id`.

keyword_clusters: project scoped. Index: `project_id`.

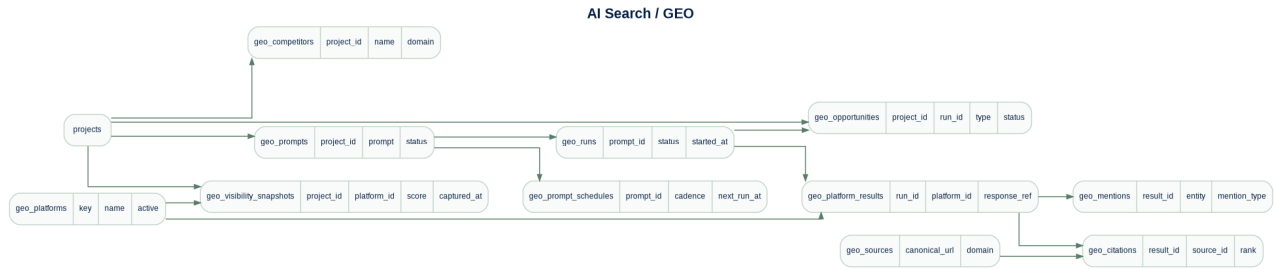
keyword_cluster_items: composite PK. Index: `keyword_id`.

serp_snapshots: append-only. Index: `keyword_id,captured_at desc`.

serp_results: `unique(snapshot_id,position,result_type,url)`. Index: `snapshot_id,position; domain`.

9. AI Search / GEO

Prompt monitoring is a first-class time-series workflow: prompt -> run -> platform result -> mentions/citations -> visibility.



PLATFORM REGISTRY

AI platform names and capabilities are backend-configured. The frontend should not hard-code platform availability.

RAW OUTPUT

Large raw model/provider responses belong in object storage; PostgreSQL stores normalized facts and a response_ref.

TRENDS

Do not compute trend arrows until at least two comparable observations exist for the same metric/platform scope.

Table	Purpose	Key fields
geo_platforms	AI platform/provider registry.	id, key, name, is_active, capabilities
geo_prompts	Saved prompt to monitor.	id, project_id, prompt, status, created_by, created_at
geo_prompt_schedules	Recurring execution schedule.	id, prompt_id, cadence, timezone, next_run_at, enabled
geo_runs	Prompt execution event.	id, prompt_id, status, trigger, started_at, completed_at...
geo_platform_results	Normalized result per platform.	id, run_id, platform_id, response_ref, response_hash, status
geo_mentions	Brand/entity mention extracted from result.	id, result_id, entity, mention_type, position_hint, sentiment
geo_sources	Canonical cited source.	id, canonical_url, domain, title, first_seen_at
geo_citations	Citation/source use in result.	id, result_id, source_id, rank, citation_text_hash
geo_competitors	Tracked competitor.	id, project_id, name, domain, status
geo_competitor_mentions	Competitor mention in result.	id, result_id, competitor_id, position_hint
geo_visibility_snapshots	Derived visibility time series.	id, project_id, platform_id, metric_key, metric_value, captured_at
geo_opportunities	Actionable GEO opportunity.	id, project_id, run_id, type, priority, status...

Integrity and query rules

geo_platforms: key unique. Index: is_active.

geo_prompts: project scoped. Index: project_id,status.

geo_prompt_schedules: one active schedule per prompt optional. Index: next_run_at where enabled.

geo_runs: append-only. Index: prompt_id,started_at desc,status.

geo_platform_results: unique(run_id,platform_id). Index: run_id,platform_id,status.

geo_mentions: result scoped. Index: result_id,entity.

geo_sources: canonical_url unique. Index: domain.

geo_citations: unique(result_id,source_id,rank). Index: result_id; source_id.

geo_competitors: unique(project_id,name). Index: project_id,status.

geo_competitor_mentions: unique(result_id,competitor_id). Index: competitor_id.

geo_visibility_snapshots: append-only derived data. Index: project_id,platform_id,captured_at desc.

geo_opportunities: project scoped. Index: project_id,status,priority.

10. Reports

Reports are generated artifacts backed by structured report records and controlled sharing.

Table	Purpose	Key fields
reports	Generated report record	id, project_id, type, status, title, period_start, period_end, created_by
report_sections	Structured report sections	id, report_id, section_key, position, payload
report_files	Generated file metadata	id, report_id, format, storage_key, size_bytes, checksum
report_schedules	Recurring report schedule	id, project_id, name, cadence, timezone, next_run_at, enabled
report_shares	Public/controlled share token	id, report_id, token_hash, expires_at, revoked_at

SHARING

Share tokens must be random, stored hashed, revocable and optionally expiring. Never expose internal report IDs as public authorization.

SCHEDULING

Scheduled reports create normal report records. Scheduling is not a separate report type; it is an execution trigger.

11. Usage and Credits

Credits are accounting data. Use an append-only ledger and derive/cache the balance.

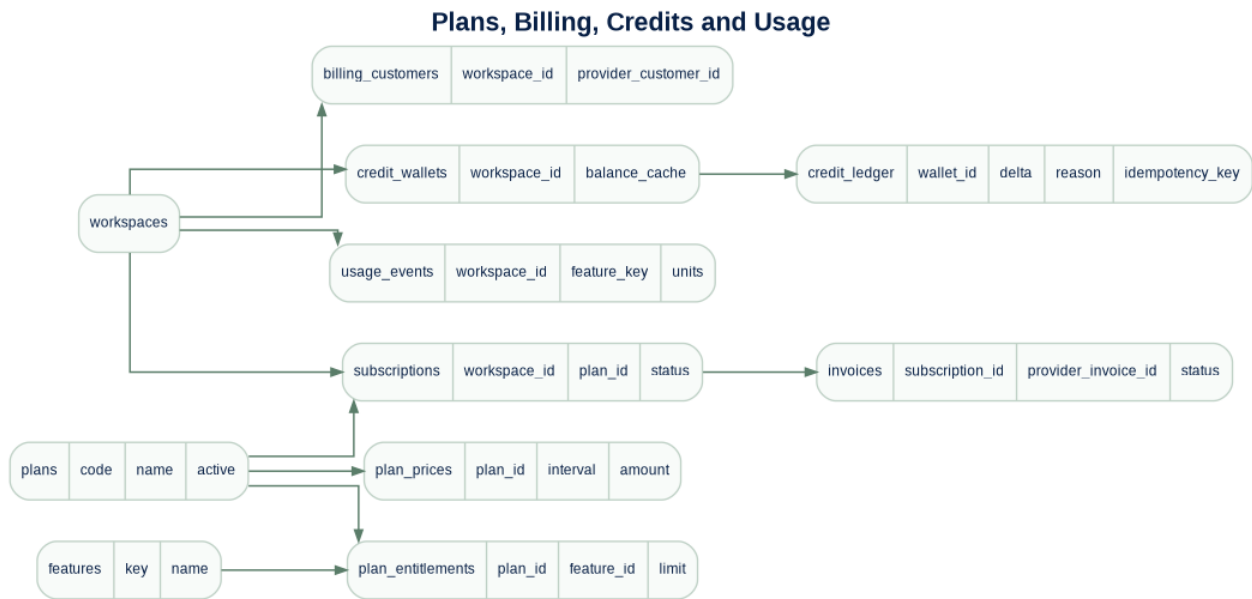


Table	Purpose	Critical rule
credit_wallets	Workspace credit account.	workspace_id unique
credit_ledger	Append-only credit movements.	idempotency_key unique; never update/delete ordinary entries
credit_purchases	Credit purchase mapping.	provider payment unique
usage_events	Billable/metered usage event.	idempotency_key unique
provider_usage	External provider consumption.	append-only telemetry

LEDGER	credit_ledger is the financial source of truth. balance_cache is a convenience and must be reconcilable from ledger entries.
IDEMPOTENCY	Every metered action should carry an idempotency key so retries cannot double-charge credits.

12. Billing, Pricing and Entitlements

Public pricing, signup, user billing and checkout should read from the same backend plan configuration.

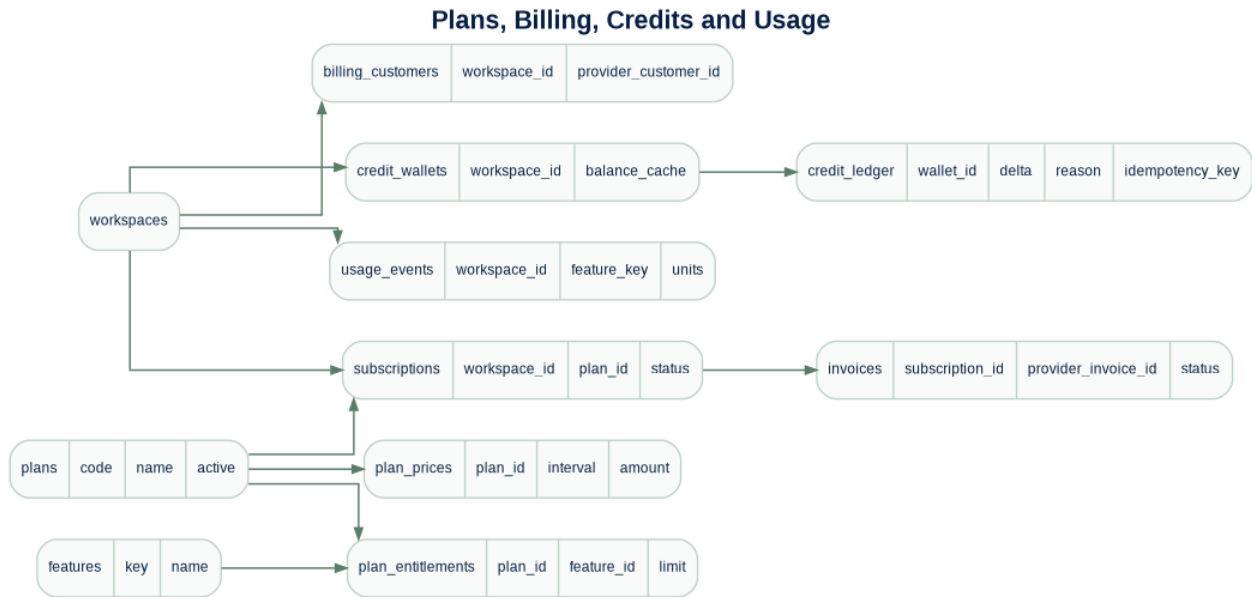


Table	Purpose	Critical rule
plans	Commercial plan definition.	code unique
plan_prices	Price per plan/interval/currency.	unique(plan_id,interval,currency,active) with partial policy
features	Entitlement feature registry.	key unique
plan_entitlements	Plan feature allowance.	unique(plan_id,feature_id)
billing_customers	Workspace-to-provider customer.	unique(provider,provider_customer_id); unique(workspace_id,provider)
subscriptions	Current/historical subscription.	provider subscription unique when present
subscription_items	Subscription price items.	unique(subscription_id,plan_price_id)
invoices	Provider invoice mirror.	provider_invoice_id unique
invoice_lines	Invoice line items.	invoice scoped
payment_events	Normalized billing/payment webhook history.	unique(provider,provider_event_id)

ONE SOURCE

Super Admin edits plan configuration; the public Pricing page reads the active plan/price rows. Stripe/provider remains payment authority, while AmpliVerify remains product-entitlement authority.

HISTORY

Changing a plan definition must not silently rewrite historical invoices or subscription periods. Preserve provider IDs and effective history.

13. Integrations and Notifications

Integrations are workspace resources; notifications are user-targeted events within a workspace.

Table	Purpose	Critical rule
integration_providers	Supported integration registry.	key unique
workspace_integrations	Workspace connection.	one/multiple per provider per design
integration_tokens	Encrypted token reference.	do not store plaintext secrets
sync_runs	Integration synchronization job.	append-only
notification_preferences	User notification settings per workspace.	unique(workspace_id,user_id,event_key,channel)
notifications	In-app notification record.	user scoped
notification_deliveries	External delivery attempt.	append attempts

SECRETS

Store OAuth tokens in a secrets system or encrypted envelope referenced by integration_tokens. Never expose tokens through normal application queries.

EVENTS

Notification events should be generated from real backend events: audit complete, scheduled report ready, low credits, billing/security notices, etc.

14. Public CMS

Blog, Guides and Help Center content are CMS-driven and publish only approved records.

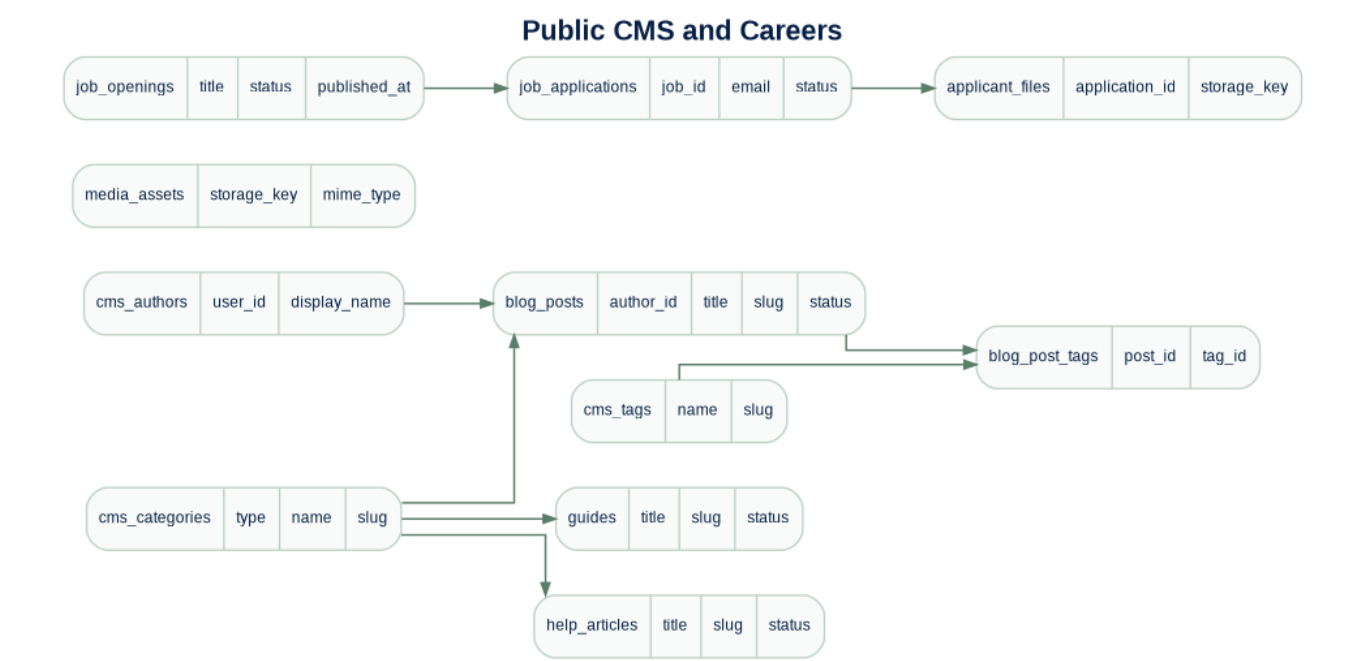


Table	Purpose	Key fields
cms_authors	Public author profile.	id, user_id, display_name, slug, bio
cms_categories	Category registry.	id, content_type, name, slug
cms_tags	Tag registry.	id, name, slug
blog_posts	Blog content.	id, author_id, category_id, title, slug, status, excerpt, body_ref, published_at
blog_post_tags	Blog/tag junction.	post_id, tag_id
guides	Guide content.	id, category_id, title, slug, status, body_ref, published_at
help_articles	Help center content.	id, category_id, title, slug, status, body_ref, published_at
media_assets	Public/CMS media metadata.	id, storage_key, mime_type, size_bytes, alt_text, uploaded_by

PUBLISH

The public site should query only status=PUBLISHED and published_at <= now. Draft content remains inaccessible by public routes.

15. Careers and Job Applications

Public Careers reads job_openings; applications are private recruitment records.

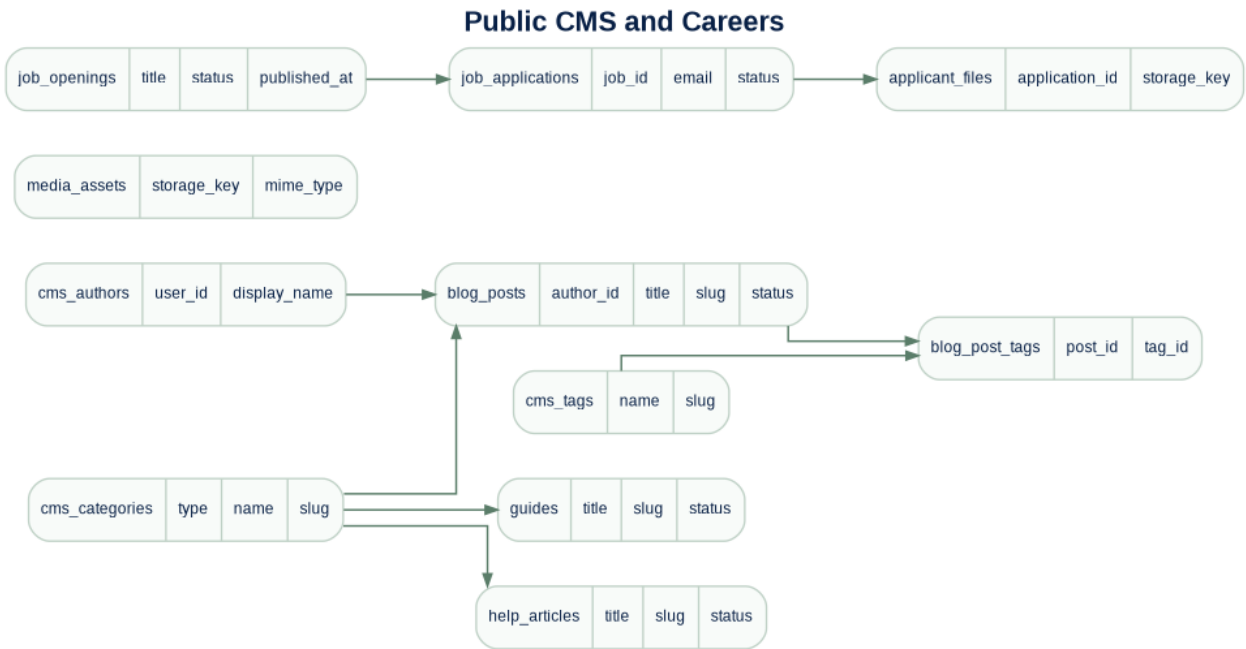


Table	Purpose	Critical rule
job_openings	Published job definition.	slug unique
job_applications	Applicant submission.	job must be open at submission; server validated
applicant_files	Private resume/application files.	private storage; no public URL
application_events	Application status history.	append-only
hiring_notes	Internal hiring notes.	strict permission scope

APPLICATION

A job application transaction must verify that the job is still OPEN/PUBLISHED at submission time; frontend visibility alone is insufficient.

FILES

Applicant files use private object storage, MIME validation, size limits and malware scanning. Do not store resumes as public media assets.

16. Super Admin, Security and Audit

Super Admin changes are consequential configuration events and require authoritative audit history.

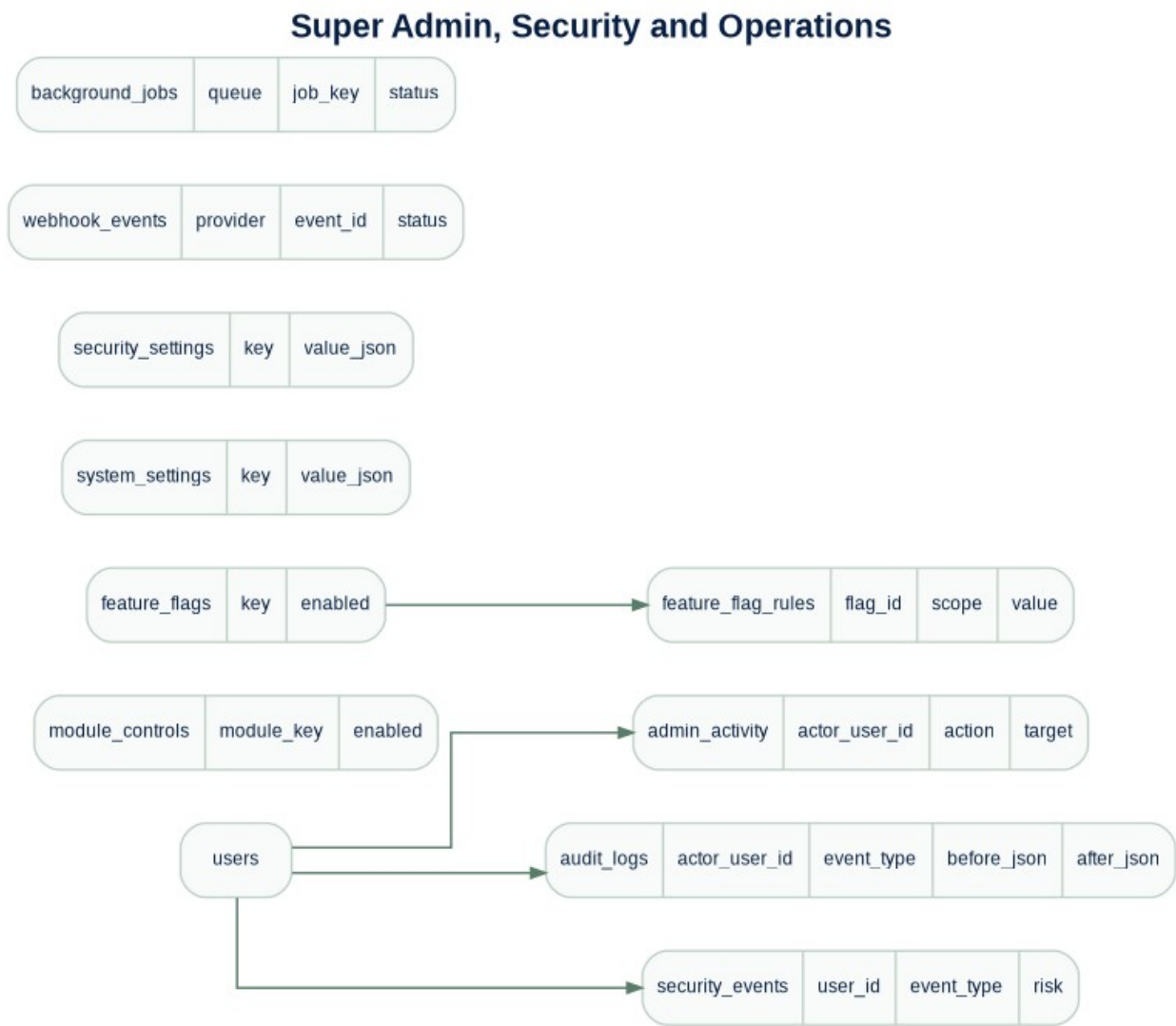


Table	Purpose	Critical rule
module_controls	Global module enable/disable.	module_key registry enforced
feature_flags	Operational flag.	unique(key,environment)
feature_flag_rules	Flag rollout targeting.	rules evaluated by priority
system_settings	General platform configuration.	allowlisted keys only
security_settings	Security configuration.	security-sensitive changes audited
admin_activity	Human-readable admin activity feed.	append-only
audit_logs	Authoritative config/security audit trail.	append-only; no ordinary deletes
security_events	Authentication/security event stream.	append-only

AUDIT

admin_activity is optimized for readable operational history; audit_logs is the authoritative before/after evidence trail. Both are append-only from ordinary application code.

SEPARATION

Module controls, feature flags, RBAC and plan entitlements are different mechanisms. Do not merge them into one table.

17. Operations, Webhooks and Background Work

Table	Purpose	Critical rule
webhook_events	Inbound webhook inbox.	unique(provider,provider_event_id)
background_jobs	Durable job metadata if needed beyond queue.	job_key unique where idempotent
provider_events	Outbound/provider operation log.	operational telemetry
system_incidents	Confirmed operational incident.	only real incidents
health_check_results	Health probe history.	short retention recommended
idempotency_keys	Request idempotency registry.	unique key

WEBHOOK

Webhook ingestion pattern: verify signature -> insert webhook_events with unique(provider,event_id) -> acknowledge -> process asynchronously -> mark processed. Duplicate deliveries become no-ops.

VOLUME

Health and provider telemetry can have shorter retention than customer records. Partition or prune high-volume operational tables.

18. Index and Performance Strategy

Workload	Recommended index / strategy
Tenant lookup	B-tree on workspace_id for every high-volume tenant table.
Project screens	Composite (project_id, status, created_at DESC).
Audit history	(project_id, started_at DESC); findings by audit_page_id + severity/status.
Keyword time series	(keyword_id, captured_at DESC); consider partitioning snapshots when volume grows.
GEO history	(prompt_id, started_at DESC), (project_id, platform_id, captured_at DESC).
Notifications	Partial index on (user_id, created_at DESC) WHERE read_at IS NULL.
Schedules	Partial index on next_run_at WHERE enabled=true.
Webhook inbox	Unique(provider,provider_event_id) plus (status,received_at).
Ledger	(wallet_id, created_at DESC) and unique idempotency_key.
JSONB	GIN only for fields actually queried; avoid blanket GIN indexes.
Search	PostgreSQL FTS/trigram for CMS/help search if needed; do not add Elastic/OpenSearch until justified.
Large event tables	Monthly partitioning only after measured need; do not over-partition early.

Query pattern rule

- Prefer composite indexes matching actual WHERE + ORDER BY patterns.
- Avoid N+1 queries for dashboard aggregates; use grouped queries or materialized/derived summaries where justified.
- Do not index every foreign key blindly if table cardinality is tiny, but index foreign keys used for joins in high-volume tables.
- Use EXPLAIN (ANALYZE, BUFFERS) before adding specialized indexes.

19. Retention, Deletion and Privacy

Data class	Recommended policy direction
Workspace/project data	Retain while account is active; deletion workflow should cascade intentionally after grace/recovery policy.
Audit/GEO/keyword history	Customer-controlled retention by plan/policy; preserve enough history for product features.
Credit ledger/invoices/payment events	Retain according to financial/legal requirements; do not hard-delete casually.
Auth/security events	Time-bounded retention appropriate to security investigations and policy.
Sessions/tokens	Delete or revoke promptly after expiry; never keep raw secrets longer than needed.
Applicant data	Define recruitment retention window; delete or anonymize after policy period.
CMS	Draft/published/archive lifecycle; media cleanup only when no references remain.
Health/provider telemetry	Short retention, aggregated longer only if useful.
Object storage	Delete artifacts when owning database record is permanently deleted, using async cleanup jobs.

DELETION

Account deletion must not be implemented as a single blind CASCADE across financial, security and audit history. Use a policy-aware deletion/anonymization workflow.

20. Migration and Implementation Order

Phase	Implement	Why this order
1	users, auth adapter, workspaces, memberships, roles, permissions	Establish identity and tenant boundary first.
2	projects, domains, pages, settings	Everything in the user app depends on project ownership.
3	plans, prices, entitlements, subscriptions, wallets, ledger	Allows signup/plan gating and usage metering.
4	audit runs, findings, optimization tasks	Core V1 SEO workflow.
5	editor, content strategy, keyword research	Builds on projects and usage.
6	GEO prompts/runs/results/citations	Higher-cost workflow; use credits from day one.
7	reports, schedules, shares, notifications	Cross-module reporting and delivery.
8	integrations and sync state	Connect external data with stable tenant model.
9	CMS and Careers	Public content/admin publishing.
10	feature flags, module controls, audit/security/ops	Operational maturity and governance.

Migration rules

- Use Prisma migrations in source control; production migrations run as a controlled deployment step.
- Prefer expand -> backfill -> switch -> contract for breaking schema changes.
- Never rename/drop columns in the same deployment that application code stops reading them unless deployment is atomic and verified.
- Backfills should be resumable and observable.
- Every external webhook and credit-consuming operation must be idempotent before launch.

21. Prisma Mapping Rules

Concern	Prisma direction
IDs	String @id @default(uuid()) or consistent CUID2 across all application tables.
PostgreSQL table names	Use @@map("snake_case") if model names are PascalCase.
Column names	Use @map("snake_case") where desired; consistency matters more than style.
Enums	Prisma enums mapped to PostgreSQL enums when stable; use text + CHECK for highly dynamic registries.
Decimal money	Store currency amounts as integer minor units (BigInt) for money; numeric/Decimal for credits/units.
JSON	Use Json only for flexible payloads, not as a substitute for relational fields used in filters/joins.
Timestamps	DateTime @db.Timestamptz(6).
Soft delete	deletedAt DateTime? plus query middleware/service rules; not every table needs it.
Secrets	Only encrypted references in DB. Secret material belongs in approved secret storage.
Relations	Use explicit onDelete behavior. RESTRICT for financial/audit history; CASCADE for pure child records where safe.

BOUNDARY

Prisma schema should encode relational integrity, but application services still enforce tenant authorization and workflow rules that cannot be represented as foreign keys.

22. Complete Table Catalog

All proposed logical tables in one implementation checklist.

Table	Purpose	Integrity	Primary index
users	Canonical application identity.	email unique; deleted users retained only per policy	email; status
auth_accounts	OAuth/provider account mapping.	unique(provider, provider_account_id)	user_id
auth_sessions	Server-side session metadata.	token_hash unique; never store raw session token	user_id, expires_at
mfa_factors	MFA enrollment metadata.	one active factor per type unless product supports multiples	user_id
passkeys	WebAuthn/passkey credentials.	credential_id unique	user_id
workspaces	Tenant root.	slug unique	status
workspace_memberships	User-to-workspace membership.	unique(workspace_id,user_id)	workspace_id; user_id; status
roles	System or custom role.	unique workspace/name for custom roles	workspace_id; is_system
permissions	Permission registry.	key unique	key
role_permissions	Role permission junction.	composite PK(role_id,permission_id)	permission_id
membership_roles	Membership role junction.	composite PK	role_id
projects	Top-level SEO project within a workspace	workspace-scoped name may repeat unless product forbids	workspace_id,status
domains	Domain/site attached to a project	unique(project_id,host)	project_id,host
pages	Known page URL	unique(domain_id,url)	domain_id,last_seen_at
project_settings	Project defaults and crawl policy	one row per project	project_id
project_data_sources	Project-to-integration binding	unique(project_id,workspace_integration_id,source_type)	project_id,status
audit_runs	One execution of an SEO audit.	immutable execution identity	project_id,status,started_at desc
audit_pages	Per-page result within a run.	unique(audit_run_id,page_id)	audit_run_id,page_id
audit_findings	Normalized issue instance.	unique(audit_page_id,fingerprint)	audit_page_id,severity,status,rule_key
audit_recommendations	Action guidance for finding.	one or many recommendations per finding	finding_id,priority
finding_events	Lifecycle history for a finding.	append-only	finding_id,created_at
optimization_tasks	Actionable work item.	tenant-bound through project	project_id,status,assigned_to
verification_runs	Verification after implementation.	append-only verification history	task_id,created_at desc
editor_documents	Editable SEO document/page workspace.	one active document per page optional, not mandatory	project_id,page_id,status
editor_versions	Immutable document snapshots.	unique(document_id,version_no)	document_id,version_no desc
editor_suggestions	AI/rule suggestions against version.	never overwrite accepted/rejected history	version_id,status,type
editor_change_events	Audit trail for editor actions.	append-only	document_id,created_at
content_ideas	Content opportunity/idea.	project scoped	project_id,status
content_briefs	Structured brief.	project scoped	project_id,status
content_plans	Editorial/content plan.	project scoped	project_id,status
content_plan_items	Plan item.	plan scoped	plan_id,status,target_date
keyword_queries	Research request metadata.	request identity immutable	project_id,created_at
keywords	Normalized keyword dictionary.	unique(normalized_term,locale,country_code)	normalized_term
keyword_metric_snapshots	Time-series keyword metrics.	append-only snapshots	keyword_id,captured_at desc
keyword_lists	Saved list.	unique(project_id,name) optional	project_id
keyword_list_items	List membership.	composite PK	keyword_id
keyword_clusters	Keyword cluster.	project scoped	project_id
keyword_cluster_items	Cluster membership.	composite PK	keyword_id
serp_snapshots	SERP observation.	append-only	keyword_id,captured_at desc
serp_results	Ranked result row.	unique(snapshot_id,position,result_type,url)	snapshot_id,position; domain
geo_platforms	AI platform/provider registry.	key unique	is_active
geo_prompts	Saved prompt to monitor.	project scoped	project_id,status
geo_prompt_schedules	Recurring execution schedule.	one active schedule per prompt optional	next_run_at where enabled
geo_runs	Prompt execution event.	append-only	prompt_id,started_at desc,status
geo_platform_results	Normalized result per platform.	unique(run_id,platform_id)	run_id,platform_id,status
geo_mentions	Brand/entity mention extracted from result.	result scoped	result_id,entity
geo_sources	Canonical cited source.	canonical_url unique	domain
geo_citations	Citation/source use in result.	unique(result_id,source_id,rank)	result_id; source_id
geo_competitors	Tracked competitor.	unique(project_id,name)	project_id,status
geo_competitor_mentions	Competitor mention in result.	unique(result_id,competitor_id)	competitor_id
geo_visibility_snapshots	Derived visibility time series.	append-only derived data	project_id,platform_id,captured_at desc
geo_opportunities	Actionable GEO opportunity.	project scoped	project_id,status,priority
reports	Generated report record	project scoped	project_id,status,created_at
report_sections	Structured report sections	unique(report_id,section_key)	report_id,position
report_files	Generated file metadata	storage_key unique	report_id
report_schedules	Recurring report schedule	project scoped	next_run_at where enabled
report_shares	Public/controlled share token	token_hash unique	report_id,expires_at
plans	Commercial plan definition.	code unique	status,display_order
plan_prices	Price per plan/interval/currency.	unique(plan_id,interval,currency,active) with partial policy	plan_id,active
features	Entitlement feature registry.	key unique	key
plan_entitlements	Plan feature allowance.	unique(plan_id,feature_id)	plan_id,feature_id
billing_customers	Workspace-to-provider customer.	unique(provider,provider_customer_id); unique(workspace_id,provider)	workspace_id
subscriptions	Current/historical subscription.	provider subscription unique when present	workspace_id,status
subscription_items	Subscription price items.	unique(subscription_id,plan_price_id)	subscription_id
invoices	Provider invoice mirror.	provider_invoice_id unique	workspace_id,status,issued_at desc
invoice_lines	Invoice line items.	invoice scoped	invoice_id
payment_events	Normalized billing/payment webhook history.	unique(provider,provider_event_id)	received_at,event_type
credit_wallets	Workspace credit account.	workspace_id unique	workspace_id
credit_ledger	Append-only credit movements.	idempotency_key unique; never update/delete ordinary entries	wallet_id,created_at desc; reference_type/reference_id
credit_purchases	Credit purchase mapping.	provider payment unique	workspace_id,status
usage_events	Billable/metered usage event.	idempotency_key unique	workspace_id,feature_key,occurred_at desc
provider_usage	External provider consumption.	append-only telemetry	provider,operation
integration_providers	Supported integration registry.	key unique	active
workspace_integrations	Workspace connection.	one/multiple per provider per design	workspace_id,provider_id,status

integration_tokens	Encrypted token reference.	do not store plaintext secrets	workspace_integration_id
sync_runs	Integration synchronization job.	append-only	workspace_integration_id,started_at desc
notification_preferences	User notification settings per workspace.	unique(workspace_id,user_id,event_key,channel)	user_id,workspace_id
notifications	In-app notification record.	user scoped	user_id,read_at,created_at desc
notification_deliveries	External delivery attempt.	append attempts	notification_id,attempted_at
cms_authors	Public author profile.	slug unique	slug
cms_categories	Category registry.	unique(content_type,slug)	content_type,slug
cms_tags	Tag registry.	slug unique	slug
blog_posts	Blog content.	slug unique	status,published_at desc
blog_post_tags	Blog/tag junction.	composite PK	tag_id
guides	Guide content.	slug unique	status,published_at desc
help_articles	Help center content.	slug unique	status,published_at desc
media_assets	Public/CMS media metadata.	storage_key unique	mime_type,uploaded_by
job_openings	Published job definition.	slug unique	status,published_at desc
job_applications	Applicant submission.	job must be open at submission; server validated	job_id,status,submitted_at desc; email
applicant_files	Private resume/application files.	private storage; no public URL	application_id
application_events	Application status history.	append-only	application_id,created_at
hiring_notes	Internal hiring notes.	strict permission scope	application_id,created_at
module_controls	Global module enable/disable.	module_key registry enforced	enabled
feature_flags	Operational flag.	unique(key,environment)	enabled,environment
feature_flag_rules	Flag rollout targeting.	rules evaluated by priority	flag_id,priority
system_settings	General platform configuration.	allowlisted keys only	key
security_settings	Security configuration.	security-sensitive changes audited	key
admin_activity	Human-readable admin activity feed.	append-only	actor_user_id,created_at desc; target_type,target_id
audit_logs	Authoritative config/security audit trail.	append-only; no ordinary deletes	created_at desc; actor_user_id; target_type,target_id
security_events	Authentication/security event stream.	append-only	user_id,created_at desc; risk_level
webhook_events	Inbound webhook inbox.	unique(provider,provider_event_id)	status,received_at
background_jobs	Durable job metadata if needed beyond queue.	job_key unique where idempotent	queue,status,run_at
provider_events	Outbound/provider operation log.	operational telemetry	provider,operation,created_at desc
system_incidents	Confirmed operational incident.	only real incidents	status,started_at desc
health_check_results	Health probe history.	short retention recommended	component_key,checked_at desc
idempotency_keys	Request idempotency registry.	unique key	expires_at

22A. Table-by-Table Field Specification

Field types are proposed PostgreSQL types. Adjust only when implementation requirements justify a change.

users

Canonical application identity.

Field	Type	Notes
id	uuid PK	
email	citext	unique normalized email
display_name	text	nullable
status	user_status	
created_at	timestampz	
updated_at	timestampz	
deleted_at	timestampz	nullable

Integrity: email unique; deleted users retained only per policy. **Index:** email; status.

auth_accounts

OAuth/provider account mapping.

Field	Type	Notes
id	uuid PK	
user_id	uuid FK users	
provider	text	
provider_account_id	text	
created_at	timestampz	

Integrity: unique(provider, provider_account_id). **Index:** user_id.

auth_sessions

Server-side session metadata.

Field	Type	Notes
id	uuid PK	
user_id	uuid FK users	
token_hash	text	
expires_at	timestampz	
ip_hash	text	nullable
user_agent	text	nullable
revoked_at	timestampz	nullable

Integrity: token_hash unique; never store raw session token. **Index:** user_id, expires_at.

mfa_factors

MFA enrollment metadata.

Field	Type	Notes
id	uuid PK	
user_id	uuid FK users	
type	mfa_type	
secret_ref	text	encrypted reference
verified_at	timestampz	nullable

Integrity: one active factor per type unless product supports multiples. **Index:** user_id.

passkeys

WebAuthn/passkey credentials.

Field	Type	Notes
id	uuid PK	
user_id	uuid FK users	
credential_id	text	
public_key	bytea	
counter	bigint	
transports	jsonb	

Integrity: credential_id unique. **Index:** user_id.

workspaces

Tenant root.

Field	Type	Notes
id	uuid PK	
name	text	
slug	citext	
status	workspace_status	
created_by	uuid FK users	
created_at	timestampz	

Integrity: slug unique. **Index:** status.

workspace_memberships

User-to-workspace membership.

Field	Type	Notes
id	uuid PK	
workspace_id	uuid FK workspaces	
user_id	uuid FK users	
status	membership_status	
joined_at	timestampz	

Integrity: unique(workspace_id,user_id). **Index:** workspace_id; user_id; status.

roles

System or custom role.

Field	Type	Notes
id	uuid PK	
workspace_id	uuid	nullable for system roles
name	text	
scope	role_scope	
is_system	boolean	

Integrity: unique workspace/name for custom roles. **Index:** workspace_id; is_system.

permissions

Permission registry.

Field	Type	Notes
id	uuid PK	
key	text	
description	text	

Integrity: key unique. **Index:** key.

role_permissions

Role permission junction.

Field	Type	Notes
role_id	uuid FK roles	
permission_id	uuid FK permissions	

Integrity: composite PK(role_id,permission_id). **Index:** permission_id.

membership_roles

Membership role junction.

Field	Type	Notes
membership_id	uuid FK memberships	
role_id	uuid FK roles	

Integrity: composite PK. **Index:** role_id.

projects

Top-level SEO project within a workspace

Field	Type	Notes
id	uuid PK	
workspace_id	uuid FK workspaces	
name	text	
status	project_status	
created_by	uuid FK users	
created_at	timestampz	
updated_at	timestampz	

Integrity: workspace-scoped name may repeat unless product forbids. **Index:** workspace_id,status.

domains

Domain/site attached to a project

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
host	citext	
canonical_url	text	
verification_status	verification_status	
created_at	timestampz	

Integrity: unique(project_id,host). **Index:** project_id,host.

pages

Known page URL

Field	Type	Notes
id	uuid PK	
domain_id	uuid FK domains	
url	text	
canonical_url	text	nullable
http_status	int	nullable
last_seen_at	timestampz	nullable
content_hash	text	nullable

Integrity: unique(domain_id,url). **Index:** domain_id,last_seen_at.

project_settings

Project defaults and crawl policy

Field	Type	Notes
project_id	uuid PK/FK projects	
locale	text	
timezone	text	
crawl_config	jsonb	
seo_defaults	jsonb	
updated_at	timestampz	

Integrity: one row per project. **Index:** project_id.

project_data_sources

Project-to-integration binding

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
workspace_integration_id	uuid FK workspace_integrations	
source_type	text	
status	connection_status	

Integrity: unique(project_id,workspace_integration_id,source_type). **Index:** project_id,status.

audit_runs

One execution of an SEO audit.

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
domain_id	uuid FK domains	nullable
status	run_status	
trigger	run_trigger	
started_at	timestamptz	
completed_at	timestamptz	nullable
summary_json	jsonb	

Integrity: immutable execution identity. **Index:** project_id,status,started_at desc.

audit_pages

Per-page result within a run.

Field	Type	Notes
id	uuid PK	
audit_run_id	uuid FK audit_runs	
page_id	uuid FK pages	
score	numeric	nullable
metrics_json	jsonb	

Integrity: unique(audit_run_id,page_id). **Index:** audit_run_id,page_id.

audit_findings

Normalized issue instance.

Field	Type	Notes
id	uuid PK	
audit_page_id	uuid FK audit_pages	
rule_key	text	
severity	finding_severity	
status	finding_status	
fingerprint	text	
details_json	jsonb	

Integrity: unique(audit_page_id,fingerprint). **Index:** audit_page_id,severity,status,rule_key.

audit_recommendations

Action guidance for finding.

Field	Type	Notes
id	uuid PK	
finding_id	uuid FK audit_findings	
priority	smallint	
title	text	
guidance	text	
source	recommendation_source	

Integrity: one or many recommendations per finding. **Index:** finding_id,priority.

finding_events

Lifecycle history for a finding.

Field	Type	Notes
id	uuid PK	
finding_id	uuid FK audit_findings	
actor_user_id	uuid	nullable
event_type	text	
payload	jsonb	
created_at	timestampz	

Integrity: append-only. **Index:** finding_id,created_at.

optimization_tasks

Actionable work item.

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
finding_id	uuid	nullable
assigned_to	uuid FK users	nullable
status	task_status	
priority	smallint	
due_at	timestampz	nullable

Integrity: tenant-bound through project. **Index:** project_id,status,assigned_to.

verification_runs

Verification after implementation.

Field	Type	Notes
id	uuid PK	
task_id	uuid FK optimization_tasks	
audit_run_id	uuid	nullable
status	run_status	
result	verification_result	
evidence_json	jsonb	
created_at	timestampz	

Integrity: append-only verification history. **Index:** task_id,created_at desc.

editor_documents

Editable SEO document/page workspace.

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
page_id	uuid FK pages	nullable
title	text	
status	editor_status	
created_by	uuid FK users	

Integrity: one active document per page optional, not mandatory. **Index:** project_id,page_id,status.

editor_versions

Immutable document snapshots.

Field	Type	Notes
id	uuid PK	
document_id	uuid FK editor_documents	
version_no	int	
content_ref	text	object storage or text key
content_hash	text	
created_by	uuid FK users	
created_at	timestampz	

Integrity: unique(document_id,version_no). **Index:** document_id,version_no desc.

editor_suggestions

AI/rule suggestions against version.

Field	Type	Notes
id	uuid PK	
version_id	uuid FK editor_versions	
type	suggestion_type	
status	suggestion_status	
suggestion_json	jsonb	
model_ref	text	nullable

Integrity: never overwrite accepted/rejected history. **Index:** version_id,status,type.

editor_change_events

Audit trail for editor actions.

Field	Type	Notes
id	uuid PK	
document_id	uuid FK editor_documents	
actor_user_id	uuid FK users	
action	text	
metadata	jsonb	
created_at	timestampz	

Integrity: append-only. **Index:** document_id,created_at.

content_ideas

Content opportunity/idea.

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
title	text	
status	content_status	
source	text	
metadata	jsonb	

Integrity: project scoped. Index: project_id,status.

content_briefs

Structured brief.

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
title	text	
primary_keyword_id	uuid	nullable
brief_json	jsonb	
status	content_status	

Integrity: project scoped. Index: project_id,status.

content_plans

Editorial/content plan.

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
name	text	
status	content_status	

Integrity: project scoped. Index: project_id,status.

content_plan_items

Plan item.

Field	Type	Notes
id	uuid PK	
plan_id	uuid FK content_plans	
brief_id	uuid FK content_briefs	nullable
title	text	
status	content_status	
target_date	date	nullable

Integrity: plan scoped. **Index:** plan_id,status,target_date.

keyword_queries

Research request metadata.

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
query	text	
locale	text	
country_code	text	
created_by	uuid FK users	
created_at	timestampz	

Integrity: request identity immutable. **Index:** project_id,created_at.

keywords

Normalized keyword dictionary.

Field	Type	Notes
id	uuid PK	
normalized_term	citext	
locale	text	
country_code	text	

Integrity: unique(normalized_term,locale,country_code). **Index:** normalized_term.

keyword_metric_snapshots

Time-series keyword metrics.

Field	Type	Notes
id	uuid PK	
keyword_id	uuid FK keywords	
provider	text	
search_volume	int	nullable
difficulty	numeric	nullable
cpc	numeric	nullable
captured_at	timestampz	

Integrity: append-only snapshots. **Index:** keyword_id,captured_at desc.

keyword_lists

Saved list.

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
name	text	
created_by	uuid FK users	

Integrity: unique(project_id,name) optional. **Index:** project_id.

keyword_list_items

List membership.

Field	Type	Notes
list_id	uuid FK keyword_lists	
keyword_id	uuid FK keywords	
added_at	timestampz	

Integrity: composite PK. **Index:** keyword_id.

keyword_clusters

Keyword cluster.

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
label	text	
method	text	
created_at	timestampz	

Integrity: project scoped. Index: project_id.

keyword_cluster_items

Cluster membership.

Field	Type	Notes
cluster_id	uuid FK keyword_clusters	
keyword_id	uuid FK keywords	
score	numeric	nullable

Integrity: composite PK. Index: keyword_id.

serp_snapshots

SERP observation.

Field	Type	Notes
id	uuid PK	
keyword_id	uuid FK keywords	
provider	text	
locale	text	
captured_at	timestampz	

Integrity: append-only. Index: keyword_id,captured_at desc.

serp_results

Ranked result row.

Field	Type	Notes
id	uuid PK	
snapshot_id	uuid FK serp_snapshots	
position	int	
url	text	
domain	citext	
title	text	nullable
result_type	text	

Integrity: unique(snapshot_id,position,result_type,url). **Index:** snapshot_id,position; domain.

geo_platforms

AI platform/provider registry.

Field	Type	Notes
id	uuid PK	
key	text	
name	text	
is_active	boolean	
capabilities	jsonb	

Integrity: key unique. **Index:** is_active.

geo_prompts

Saved prompt to monitor.

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
prompt	text	
status	geo_prompt_status	
created_by	uuid FK users	
created_at	timestampz	

Integrity: project scoped. **Index:** project_id,status.

geo_prompt_schedules

Recurring execution schedule.

Field	Type	Notes
id	uuid PK	
prompt_id	uuid FK geo_prompts	
cadence	text	
timezone	text	
next_run_at	timestampz	
enabled	boolean	

Integrity: one active schedule per prompt optional. **Index:** next_run_at where enabled.

geo_runs

Prompt execution event.

Field	Type	Notes
id	uuid PK	
prompt_id	uuid FK geo_prompts	
status	run_status	
trigger	run_trigger	
started_at	timestampz	
completed_at	timestampz	nullable
credit_cost	numeric	nullable

Integrity: append-only. **Index:** prompt_id,started_at desc,status.

geo_platform_results

Normalized result per platform.

Field	Type	Notes
id	uuid PK	
run_id	uuid FK geo_runs	
platform_id	uuid FK geo_platforms	
response_ref	text	object storage ref
response_hash	text	nullable
status	result_status	

Integrity: unique(run_id,platform_id). **Index:** run_id,platform_id,status.

geo_mentions

Brand/entity mention extracted from result.

Field	Type	Notes
id	uuid PK	
result_id	uuid FK geo_platform_results	
entity	text	
mention_type	text	
position_hint	int	nullable
sentiment	text	nullable

Integrity: result scoped. Index: result_id,entity.

geo_sources

Canonical cited source.

Field	Type	Notes
id	uuid PK	
canonical_url	text	
domain	citext	
title	text	nullable
first_seen_at	timestampz	

Integrity: canonical_url unique. Index: domain.

geo_citations

Citation/source use in result.

Field	Type	Notes
id	uuid PK	
result_id	uuid FK geo_platform_results	
source_id	uuid FK geo_sources	
rank	int	nullable
citation_text_hash	text	nullable

Integrity: unique(result_id,source_id,rank). Index: result_id; source_id.

geo_competitors

Tracked competitor.

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
name	text	
domain	citext	nullable
status	text	

Integrity: unique(project_id,name). **Index:** project_id,status.

geo_competitor_mentions

Competitor mention in result.

Field	Type	Notes
id	uuid PK	
result_id	uuid FK geo_platform_results	
competitor_id	uuid FK geo_competitors	
position_hint	int	nullable

Integrity: unique(result_id,competitor_id). **Index:** competitor_id.

geo_visibility_snapshots

Derived visibility time series.

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
platform_id	uuid FK geo_platforms	
metric_key	text	
metric_value	numeric	
captured_at	timestampz	

Integrity: append-only derived data. **Index:** project_id,platform_id,captured_at desc.

geo_opportunities

Actionable GEO opportunity.

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
run_id	uuid FK geo_runs	nullable
type	text	
priority	smallint	
status	task_status	
details_json	jsonb	

Integrity: project scoped. **Index:** project_id,status,priority.

reports

Generated report record

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
type	report_type	
status	run_status	
title	text	
period_start	date	nullable
period_end	date	nullable
created_by	uuid FK users	

Integrity: project scoped. **Index:** project_id,status,created_at.

report_sections

Structured report sections

Field	Type	Notes
id	uuid PK	
report_id	uuid FK reports	
section_key	text	
position	int	
payload	jsonb	

Integrity: unique(report_id,section_key). **Index:** report_id,position.

report_files

Generated file metadata

Field	Type	Notes
id	uuid PK	
report_id	uuid FK reports	
format	text	
storage_key	text	
size_bytes	bigint	
checksum	text	

Integrity: storage_key unique. Index: report_id.

report_schedules

Recurring report schedule

Field	Type	Notes
id	uuid PK	
project_id	uuid FK projects	
name	text	
cadence	text	
timezone	text	
next_run_at	timestamptz	
enabled	boolean	

Integrity: project scoped. Index: next_run_at where enabled.

report_shares

Public/controlled share token

Field	Type	Notes
id	uuid PK	
report_id	uuid FK reports	
token_hash	text	
expires_at	timestamptz	nullable
revoked_at	timestamptz	nullable

Integrity: token_hash unique. Index: report_id,expires_at.

plans

Commercial plan definition.

Field	Type	Notes
id	uuid PK	
code	text	
name	text	
status	plan_status	
display_order	int	

Integrity: code unique. **Index:** status,display_order.

plan_prices

Price per plan/interval/currency.

Field	Type	Notes
id	uuid PK	
plan_id	uuid FK plans	
billing_interval	billing_interval	
currency	char(3)	
amount_minor	bigint	
provider_price_id	text	nullable
active	boolean	

Integrity: unique(plan_id,interval,currency,active) with partial policy. **Index:** plan_id,active.

features

Entitlement feature registry.

Field	Type	Notes
id	uuid PK	
key	text	
name	text	
value_type	entitlement_value_type	

Integrity: key unique. **Index:** key.

plan_entitlements

Plan feature allowance.

Field	Type	Notes
id	uuid PK	
plan_id	uuid FK plans	
feature_id	uuid FK features	
enabled	boolean	
limit_numeric	numeric	nullable
config_json	jsonb	

Integrity: unique(plan_id,feature_id). **Index:** plan_id,feature_id.

billing_customers

Workspace-to-provider customer.

Field	Type	Notes
id	uuid PK	
workspace_id	uuid FK workspaces	
provider	text	
provider_customer_id	text	

Integrity: unique(provider,provider_customer_id); unique(workspace_id,provider). **Index:** workspace_id.

subscriptions

Current/historical subscription.

Field	Type	Notes
id	uuid PK	
workspace_id	uuid FK workspaces	
plan_id	uuid FK plans	
provider_subscription_id	text	nullable
status	subscription_status	
current_period_start	timestampz	
current_period_end	timestampz	
cancel_at_period_end	boolean	

Integrity: provider subscription unique when present. **Index:** workspace_id,status.

subscription_items

Subscription price items.

Field	Type	Notes
id	uuid PK	
subscription_id	uuid FK subscriptions	
plan_price_id	uuid FK plan_prices	
quantity	int	

Integrity: unique(subscription_id,plan_price_id). Index: subscription_id.

invoices

Provider invoice mirror.

Field	Type	Notes
id	uuid PK	
subscription_id	uuid FK subscriptions	nullable
workspace_id	uuid FK workspaces	
provider_invoice_id	text	
status	invoice_status	
currency	char(3)	
total_minor	bigint	
issued_at	timestampz	nullable

Integrity: provider_invoice_id unique. Index: workspace_id,status,issued_at desc.

invoice_lines

Invoice line items.

Field	Type	Notes
id	uuid PK	
invoice_id	uuid FK invoices	
description	text	
amount_minor	bigint	
quantity	numeric	nullable

Integrity: invoice scoped. Index: invoice_id.

payment_events

Normalized billing/payment webhook history.

Field	Type	Notes
id	uuid PK	
workspace_id	uuid	nullable
provider	text	
provider_event_id	text	
event_type	text	
payload_ref	text	nullable
received_at	timestampz	

Integrity: unique(provider,provider_event_id). **Index:** received_at,event_type.

credit_wallets

Workspace credit account.

Field	Type	Notes
id	uuid PK	
workspace_id	uuid FK workspaces	
balance_cache	numeric	
updated_at	timestampz	

Integrity: workspace_id unique. **Index:** workspace_id.

credit_ledger

Append-only credit movements.

Field	Type	Notes
id	uuid PK	
wallet_id	uuid FK credit_wallets	
delta	numeric	
reason	credit_reason	
reference_type	text	nullable
reference_id	uuid	nullable
idempotency_key	text	
created_by	uuid	nullable
created_at	timestampz	

Integrity: idempotency_key unique; never update/delete ordinary entries. **Index:** wallet_id,created_at desc; reference_type/reference_id.

credit_purchases

Credit purchase mapping.

Field	Type	Notes
id	uuid PK	
workspace_id	uuid FK workspaces	
provider_payment_id	text	
credits	numeric	
amount_minor	bigint	
status	payment_status	

Integrity: provider payment unique. Index: workspace_id,status.

usage_events

Billable/metered usage event.

Field	Type	Notes
id	uuid PK	
workspace_id	uuid FK workspaces	
project_id	uuid	nullable
feature_key	text	
units	numeric	
idempotency_key	text	
occurred_at	timestampz	

Integrity: idempotency_key unique. Index: workspace_id,feature_key,occurred_at desc.

provider_usage

External provider consumption.

Field	Type	Notes
id	uuid PK	
usage_event_id	uuid FK usage_events	nullable
provider	text	
operation	text	
input_units	numeric	nullable
output_units	numeric	nullable
cost_minor	bigint	nullable
currency	char(3)	nullable

Integrity: append-only telemetry. Index: provider,operation.

integration_providers

Supported integration registry.

Field	Type	Notes
id	uuid PK	
key	text	
name	text	
auth_type	text	
active	boolean	
scopes_json	jsonb	

Integrity: key unique. **Index:** active.

workspace_integrations

Workspace connection.

Field	Type	Notes
id	uuid PK	
workspace_id	uuid FK workspaces	
provider_id	uuid FK integration_providers	
status	connection_status	
external_account_ref	text	nullable
connected_by	uuid FK users	

Integrity: one/multiple per provider per design. **Index:** workspace_id,provider_id,status.

integration_tokens

Encrypted token reference.

Field	Type	Notes
id	uuid PK	
workspace_integration_id	uuid FK workspace_integrations	
access_secret_ref	text	
refresh_secret_ref	text	nullable
expires_at	timestampz	nullable

Integrity: do not store plaintext secrets. **Index:** workspace_integration_id.

sync_runs

Integration synchronization job.

Field	Type	Notes
id	uuid PK	
workspace_integration_id	uuid FK workspace_integrations	
sync_type	text	
status	run_status	
started_at	timestampz	
completed_at	timestampz	nullable
error_code	text	nullable

Integrity: append-only. **Index:** workspace_integration_id,started_at desc.

notification_preferences

User notification settings per workspace.

Field	Type	Notes
id	uuid PK	
workspace_id	uuid FK workspaces	
user_id	uuid FK users	
event_key	text	
channel	notification_channel	
enabled	boolean	

Integrity: unique(workspace_id,user_id,event_key,channel). **Index:** user_id,workspace_id.

notifications

In-app notification record.

Field	Type	Notes
id	uuid PK	
workspace_id	uuid FK workspaces	
user_id	uuid FK users	
event_key	text	
title	text	
body	text	
read_at	timestampz	nullable
created_at	timestampz	

Integrity: user scoped. **Index:** user_id,read_at,created_at desc.

notification_deliveries

External delivery attempt.

Field	Type	Notes
id	uuid PK	
notification_id	uuid FK notifications	
channel	notification_channel	
status	delivery_status	
provider_message_id	text	nullable
attempted_at	timestampz	

Integrity: append attempts. Index: notification_id,attempted_at.

cms_authors

Public author profile.

Field	Type	Notes
id	uuid PK	
user_id	uuid FK users	nullable
display_name	text	
slug	citext	
bio	text	nullable

Integrity: slug unique. Index: slug.

cms_categories

Category registry.

Field	Type	Notes
id	uuid PK	
content_type	cms_content_type	
name	text	
slug	citext	

Integrity: unique(content_type,slug). Index: content_type,slug.

cms_tags

Tag registry.

Field	Type	Notes
id	uuid PK	
name	text	
slug	citext	

Integrity: slug unique. Index: slug.

blog_posts

Blog content.

Field	Type	Notes
id	uuid PK	
author_id	uuid FK cms_authors	nullable
category_id	uuid FK cms_categories	nullable
title	text	
slug	citext	
status	publish_status	
excerpt	text	nullable
body_ref	text	
published_at	timestamptz	nullable

Integrity: slug unique. Index: status,published_at desc.

blog_post_tags

Blog/tag junction.

Field	Type	Notes
post_id	uuid FK blog_posts	
tag_id	uuid FK cms_tags	

Integrity: composite PK. Index: tag_id.

guides

Guide content.

Field	Type	Notes
id	uuid PK	
category_id	uuid	nullable
title	text	
slug	citext	
status	publish_status	
body_ref	text	
published_at	timestampz	nullable

Integrity: slug unique. **Index:** status,published_at desc.

help_articles

Help center content.

Field	Type	Notes
id	uuid PK	
category_id	uuid	nullable
title	text	
slug	citext	
status	publish_status	
body_ref	text	
published_at	timestampz	nullable

Integrity: slug unique. **Index:** status,published_at desc.

media_assets

Public/CMS media metadata.

Field	Type	Notes
id	uuid PK	
storage_key	text	
mime_type	text	
size_bytes	bigint	
alt_text	text	nullable
uploaded_by	uuid FK users	

Integrity: storage_key unique. **Index:** mime_type,uploaded_by.

job_openings

Published job definition.

Field	Type	Notes
id	uuid PK	
title	text	
slug	citext	
status	job_status	
location_text	text	nullable
employment_type	text	
work_arrangement	text	nullable
description_ref	text	
published_at	timestamptz	nullable

Integrity: slug unique. **Index:** status,published_at desc.

job_applications

Applicant submission.

Field	Type	Notes
id	uuid PK	
job_id	uuid FK job_openings	
first_name	text	
last_name	text	
email	citext	
phone	text	nullable
linkedin_url	text	nullable
portfolio_url	text	nullable
message	text	nullable
status	application_status	
submitted_at	timestamptz	

Integrity: job must be open at submission; server validated. **Index:** job_id,status,submitted_at desc; email.

applicant_files

Private resume/application files.

Field	Type	Notes
id	uuid PK	
application_id	uuid FK job_applications	
storage_key	text	
mime_type	text	
size_bytes	bigint	
malware_scan_status	scan_status	
checksum	text	

Integrity: private storage; no public URL. **Index:** application_id.

application_events

Application status history.

Field	Type	Notes
id	uuid PK	
application_id	uuid FK job_applications	
actor_user_id	uuid	nullable
event_type	text	
metadata	jsonb	
created_at	timestampz	

Integrity: append-only. Index: application_id,created_at.

hiring_notes

Internal hiring notes.

Field	Type	Notes
id	uuid PK	
application_id	uuid FK job_applications	
author_user_id	uuid FK users	
note	text	
created_at	timestampz	

Integrity: strict permission scope. Index: application_id,created_at.

module_controls

Global module enable/disable.

Field	Type	Notes
module_key	text PK	
enabled	boolean	
updated_by	uuid FK users	
updated_at	timestampz	

Integrity: module_key registry enforced. Index: enabled.

feature_flags

Operational flag.

Field	Type	Notes
id	uuid PK	
key	text	
enabled	boolean	
description	text	
environment	text	

Integrity: unique(key,environment). **Index:** enabled,environment.

feature_flag_rules

Flag rollout targeting.

Field	Type	Notes
id	uuid PK	
flag_id	uuid FK feature_flags	
scope_type	text	
scope_value	text	
percentage	numeric	nullable
priority	int	

Integrity: rules evaluated by priority. **Index:** flag_id,priority.

system_settings

General platform configuration.

Field	Type	Notes
key	text PK	
value_json	jsonb	
updated_by	uuid FK users	
updated_at	timestampz	

Integrity: allowlisted keys only. **Index:** key.

security_settings

Security configuration.

Field	Type	Notes
key	text PK	
value_json	jsonb	
updated_by	uuid FK users	
updated_at	timestampz	

Integrity: security-sensitive changes audited. **Index:** key.

admin_activity

Human-readable admin activity feed.

Field	Type	Notes
id	uuid PK	
actor_user_id	uuid FK users	
action	text	
target_type	text	
target_id	text	nullable
summary	text	
created_at	timestampz	

Integrity: append-only. **Index:** actor_user_id,created_at desc; target_type,target_id.

audit_logs

Authoritative config/security audit trail.

Field	Type	Notes
id	uuid PK	
actor_user_id	uuid	nullable
event_type	text	
target_type	text	
target_id	text	nullable
before_json	jsonb	nullable
after_json	jsonb	nullable
request_id	text	nullable
created_at	timestampz	

Integrity: append-only; no ordinary deletes. **Index:** created_at desc; actor_user_id; target_type,target_id.

security_events

Authentication/security event stream.

Field	Type	Notes
id	uuid PK	
user_id	uuid	nullable
event_type	text	
risk_level	text	
ip_hash	text	nullable
metadata	jsonb	
created_at	timestampz	

Integrity: append-only. **Index:** user_id,created_at desc; risk_level.

webhook_events

Inbound webhook inbox.

Field	Type	Notes
id	uuid PK	
provider	text	
provider_event_id	text	
event_type	text	
payload_ref	text	
status	webhook_status	
attempt_count	int	
received_at	timestampz	
processed_at	timestampz	nullable

Integrity: unique(provider,provider_event_id). **Index:** status,received_at.

background_jobs

Durable job metadata if needed beyond queue.

Field	Type	Notes
id	uuid PK	
queue	text	
job_key	text	
status	job_run_status	
payload_ref	text	nullable
attempts	int	
run_at	timestampz	

Integrity: job_key unique where idempotent. **Index:** queue,status,run_at.

provider_events

Outbound/provider operation log.

Field	Type	Notes
id	uuid PK	
provider	text	
operation	text	
status	text	
request_id	text	nullable
latency_ms	int	nullable
created_at	timestampz	

Integrity: operational telemetry. Index: provider,operation,created_at desc.

system_incidents

Confirmed operational incident.

Field	Type	Notes
id	uuid PK	
status	incident_status	
title	text	
started_at	timestampz	
resolved_at	timestampz	nullable
public_summary	text	nullable

Integrity: only real incidents. Index: status,started_at desc.

health_check_results

Health probe history.

Field	Type	Notes
id	uuid PK	
component_key	text	
status	health_status	
latency_ms	int	nullable
checked_at	timestampz	

Integrity: short retention recommended. Index: component_key,checked_at desc.

idempotency_keys

Request idempotency registry.

Field	Type	Notes
key	text PK	
scope	text	
request_hash	text	
response_ref	text	nullable
expires_at	timestampz	

Integrity: unique key. Index: expires_at.

23. Proposed Prisma Schema Core

This appendix is a practical starting point. It intentionally focuses on structural models; Better Auth adapter-specific models should follow the installed adapter version.

```
generator client {
  provider = "prisma-client-js"
}

datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}

enum UserStatus { ACTIVE SUSPENDED DELETED }
enum WorkspaceStatus { ACTIVE SUSPENDED CLOSED }
enum MembershipStatus { INVITED ACTIVE SUSPENDED REMOVED }
enum RunStatus { QUEUED RUNNING SUCCEEDED FAILED CANCELED }
enum PublishStatus { DRAFT PUBLISHED ARCHIVED }
enum SubscriptionStatus { TRIALING ACTIVE PAST_DUE CANCELED INCOMPLETE }

model User {
  id          String @id @default(uuid()) @db.Uuid
  email       String @unique @db.Citext
  displayName String? @map("display_name")
  status      UserStatus @default(ACTIVE)
  createdAt   DateTime @default(now()) @map("created_at") @db.Timestamptz(6)
  updatedAt   DateTime @updatedAt @map("updated_at") @db.Timestamptz(6)
  memberships WorkspaceMembership[]
  @@map("users")
}

model Workspace {
  id          String @id @default(uuid()) @db.Uuid
  name        String
  slug        String @unique @db.Citext
  status      WorkspaceStatus @default(ACTIVE)
  createdBy   String @map("created_by") @db.Uuid
  createdAt   DateTime @default(now()) @map("created_at") @db.Timestamptz(6)
  memberships WorkspaceMembership[]
  projects    Project[]
  subscriptions Subscription[]
  creditWallet CreditWallet?
  @@map("workspaces")
}

model WorkspaceMembership {
  id          String @id @default(uuid()) @db.Uuid
  workspaceId String @map("workspace_id") @db.Uuid
  userId      String @map("user_id") @db.Uuid
  status      MembershipStatus @default(ACTIVE)
  joinedAt    DateTime @default(now()) @map("joined_at") @db.Timestamptz(6)
  workspace   Workspace @relation(fields:[workspaceId], references:[id], onDelete:Cascade)
  user        User @relation(fields:[userId], references:[id], onDelete:Cascade)
  roles       MembershipRole[]
  @@unique([workspaceId,userId])
  @@index([userId,status])
  @@map("workspace_memberships")
}
```



```

model Role {
  id          String @id @default(uuid()) @db.Uuid
  workspaceId String? @map("workspace_id") @db.Uuid
  name        String
  scope       String
  isSystem    Boolean @default(false) @map("is_system")
  permissions RolePermission[]
  memberships MembershipRole[]
  @@index([workspaceId])
  @@map("roles")
}

model Permission {
  id          String @id @default(uuid()) @db.Uuid
  key         String @unique
  description String
  roles       RolePermission[]
  @@map("permissions")
}

model RolePermission {
  roleId       String @map("role_id") @db.Uuid
  permissionId String @map("permission_id") @db.Uuid
  role Role @relation(fields:[roleId], references:[id], onDelete:Cascade)
  permission Permission @relation(fields:[permissionId], references:[id], onDelete:Cascade)
  @@id([roleId,permissionId])
  @@map("role_permissions")
}

model MembershipRole {
  membershipId String @map("membership_id") @db.Uuid
  roleId       String @map("role_id") @db.Uuid
  membership WorkspaceMembership @relation(fields:[membershipId], references:[id], onDelete:Cascade)
  role Role @relation(fields:[roleId], references:[id], onDelete:Restrict)
  @@id([membershipId,roleId])
  @@map("membership_roles")
}

model Project {
  id          String @id @default(uuid()) @db.Uuid
  workspaceId String @map("workspace_id") @db.Uuid
  name        String
  status      String
  createdBy   String @map("created_by") @db.Uuid
  createdAt   DateTime @default(now()) @map("created_at") @db.Timestamptz(6)
  updatedAt   DateTime @updatedAt @map("updated_at") @db.Timestamptz(6)
  workspace Workspace @relation(fields:[workspaceId], references:[id], onDelete:Cascade)
  domains Domain[]
  audits AuditRun[]
  geoPrompts GeoPrompt[]
  @@index([workspaceId,status])
  @@map("projects")
}

```

```

model Domain {
    id                String @id @default(uuid()) @db.Uuid
    projectId         String @map("project_id") @db.Uuid
    host              String @db.Citext
    canonicalUrl      String @map("canonical_url")
    verificationStatus String @map("verification_status")
    project Project @relation(fields:[projectId], references:[id], onDelete:Cascade)
    pages Page[]
    @@unique([projectId,host])
    @@map("domains")
}

model Page {
    id                String @id @default(uuid()) @db.Uuid
    domainId          String @map("domain_id") @db.Uuid
    url               String
    canonicalUrl      String? @map("canonical_url")
    httpStatus        Int? @map("http_status")
    lastSeenAt        DateTime? @map("last_seen_at") @db.Timestamptz(6)
    contentHash       String? @map("content_hash")
    domain Domain @relation(fields:[domainId], references:[id], onDelete:Cascade)
    @@unique([domainId,url])
    @@index([domainId,lastSeenAt])
    @@map("pages")
}

model AuditRun {
    id                String @id @default(uuid()) @db.Uuid
    projectId         String @map("project_id") @db.Uuid
    domainId          String? @map("domain_id") @db.Uuid
    status            RunStatus
    trigger           String
    startedAt         DateTime @default(now()) @map("started_at") @db.Timestamptz(6)
    completedAt       DateTime? @map("completed_at") @db.Timestamptz(6)
    summaryJson       Json? @map("summary_json")
    project Project @relation(fields:[projectId], references:[id], onDelete:Cascade)
    pages AuditPage[]
    @@index([projectId,startedAt(sort: Desc)])
    @@map("audit_runs")
}

model AuditPage {
    id                String @id @default(uuid()) @db.Uuid
    auditRunId        String @map("audit_run_id") @db.Uuid
    pageId            String @map("page_id") @db.Uuid
    score             Decimal? @db.Decimal(6,2)
    metricsJson       Json? @map("metrics_json")
    auditRun AuditRun @relation(fields:[auditRunId], references:[id], onDelete:Cascade)
    findings AuditFinding[]
    @@unique([auditRunId,pageId])
    @@map("audit_pages")
}

```

```

model AuditFinding {
    id          String @id @default(uuid()) @db.Uuid
    auditPageId String @map("audit_page_id") @db.Uuid
    ruleKey      String @map("rule_key")
    severity     String
    status       String
    fingerprint  String
    detailsJson  Json? @map("details_json")
    auditPage    AuditPage @relation(fields:[auditPageId], references:[id], onDelete:Cascade)
    recommendations AuditRecommendation[]
    @@unique([auditPageId, fingerprint])
    @@index([auditPageId, severity, status])
    @@map("audit_findings")
}

model AuditRecommendation {
    id          String @id @default(uuid()) @db.Uuid
    findingId    String @map("finding_id") @db.Uuid
    priority     Int
    title        String
    guidance     String
    source       String
    finding      AuditFinding @relation(fields:[findingId], references:[id], onDelete:Cascade)
    @@index([findingId, priority])
    @@map("audit_recommendations")
}

model GeoPlatform {
    id          String @id @default(uuid()) @db.Uuid
    key         String @unique
    name        String
    isActive    Boolean @default(true) @map("is_active")
    capabilities Json?
    results     GeoPlatformResult[]
    @@map("geo_platforms")
}

model GeoPrompt {
    id          String @id @default(uuid()) @db.Uuid
    projectId   String @map("project_id") @db.Uuid
    prompt      String
    status      String
    createdBy   String @map("created_by") @db.Uuid
    createdAt   DateTime @default(now()) @map("created_at") @db.Timestamptz(6)
    project     Project @relation(fields:[projectId], references:[id], onDelete:Cascade)
    runs        GeoRun[]
    @@index([projectId, status])
    @@map("geo_prompts")
}

model GeoRun {
    id          String @id @default(uuid()) @db.Uuid
    promptId    String @map("prompt_id") @db.Uuid
    status      RunStatus

```

```

trigger      String
startedAt    DateTime @default(now()) @map("started_at") @db.Timestamptz(6)
completedAt  DateTime? @map("completed_at") @db.Timestamptz(6)
creditCost   Decimal? @map("credit_cost") @db.Decimal(18,4)
prompt       GeoPrompt @relation(fields:[promptId], references:[id], onDelete:Cascade)
platformResults GeoPlatformResult[]
@@index([promptId,startedAt(sort: Desc)])
@@map("geo_runs")
}

model GeoPlatformResult {
  id          String @id @default(uuid()) @db.Uuid
  runId       String @map("run_id") @db.Uuid
  platformId  String @map("platform_id") @db.Uuid
  responseRef String @map("response_ref")
  responseHash String? @map("response_hash")
  status      String
  run GeoRun @relation(fields:[runId], references:[id], onDelete:Cascade)
  platform GeoPlatform @relation(fields:[platformId], references:[id], onDelete:Restrict)
  citations GeoCitation[]
  @@unique([runId,platformId])
  @@map("geo_platform_results")
}

model GeoSource {
  id          String @id @default(uuid()) @db.Uuid
  canonicalUrl String @unique @map("canonical_url")
  domain      String @db.Citext
  title       String?
  firstSeenAt DateTime @default(now()) @map("first_seen_at") @db.Timestamptz(6)
  citations GeoCitation[]
  @@index([domain])
  @@map("geo_sources")
}

model GeoCitation {
  id          String @id @default(uuid()) @db.Uuid
  resultId    String @map("result_id") @db.Uuid
  sourceId    String @map("source_id") @db.Uuid
  rank        Int?
  result GeoPlatformResult @relation(fields:[resultId], references:[id], onDelete:Cascade)
  source GeoSource @relation(fields:[sourceId], references:[id], onDelete:Restrict)
  @@index([resultId])
  @@index([sourceId])
  @@map("geo_citations")
}

model Plan {
  id          String @id @default(uuid()) @db.Uuid
  code        String @unique
  name        String
  status      String
  displayOrder Int @map("display_order")
  prices PlanPrice[]

```

```

    entitlements PlanEntitlement[]
    subscriptions Subscription[]
    @@index([status,displayOrder])
    @@map("plans")
}

model PlanPrice {
    id          String @id @default(uuid()) @db.Uuid
    planId      String @map("plan_id") @db.Uuid
    billingInterval String @map("billing_interval")
    currency    String @db.Char(3)
    amountMinor BigInt @map("amount_minor")
    providerPriceId String? @map("provider_price_id")
    active      Boolean @default(true)
    plan Plan @relation(fields:[planId], references:[id], onDelete:Restrict)
    @@index([planId,active])
    @@map("plan_prices")
}

model Feature {
    id          String @id @default(uuid()) @db.Uuid
    key         String @unique
    name        String
    valueType   String @map("value_type")
    entitlements PlanEntitlement[]
    @@map("features")
}

model PlanEntitlement {
    id          String @id @default(uuid()) @db.Uuid
    planId      String @map("plan_id") @db.Uuid
    featureId   String @map("feature_id") @db.Uuid
    enabled     Boolean @default(false)
    limitNumeric Decimal? @map("limit_numeric") @db.Decimal(18,4)
    configJson  Json? @map("config_json")
    plan Plan @relation(fields:[planId], references:[id], onDelete:Cascade)
    feature Feature @relation(fields:[featureId], references:[id], onDelete:Restrict)
    @@unique([planId,featureId])
    @@map("plan_entitlements")
}

model Subscription {
    id          String @id @default(uuid()) @db.Uuid
    workspaceId String @map("workspace_id") @db.Uuid
    planId      String @map("plan_id") @db.Uuid
    providerSubscriptionId String? @unique @map("provider_subscription_id")
    status      SubscriptionStatus
    currentPeriodStart DateTime @map("current_period_start") @db.Timestamptz(6)
    currentPeriodEnd   DateTime @map("current_period_end") @db.Timestamptz(6)
    cancelAtPeriodEnd Boolean @default(false) @map("cancel_at_period_end")
    workspace Workspace @relation(fields:[workspaceId], references:[id], onDelete:Restrict)
    plan Plan @relation(fields:[planId], references:[id], onDelete:Restrict)
    @@index([workspaceId,status])
    @@map("subscriptions")
}

```

```

}

model CreditWallet {
    id String @id @default(uuid()) @db.Uuid
    workspaceId String @unique @map("workspace_id") @db.Uuid
    balanceCache Decimal @default(0) @map("balance_cache") @db.Decimal(18,4)
    updatedAt DateTime @updatedAt @map("updated_at") @db.Timestamptz(6)
    workspace Workspace @relation(fields:[workspaceId], references:[id], onDelete:Restrict)
    entries CreditLedger[]
    @@map("credit_wallets")
}

model CreditLedger {
    id String @id @default(uuid()) @db.Uuid
    walletId String @map("wallet_id") @db.Uuid
    delta Decimal @db.Decimal(18,4)
    reason String
    referenceType String? @map("reference_type")
    referenceId String? @map("reference_id") @db.Uuid
    idempotencyKey String @unique @map("idempotency_key")
    createdBy String? @map("created_by") @db.Uuid
    createdAt DateTime @default(now()) @map("created_at") @db.Timestamptz(6)
    wallet CreditWallet @relation(fields:[walletId], references:[id], onDelete:Restrict)
    @@index([walletId,createdAt(sort: Desc)])
    @@map("credit_ledger")
}

model BlogPost {
    id String @id @default(uuid()) @db.Uuid
    title String
    slug String @unique @db.Citext
    status PublishStatus @default(DRAFT)
    excerpt String?
    bodyRef String @map("body_ref")
    publishedAt DateTime? @map("published_at") @db.Timestamptz(6)
    @@index([status,publishedAt(sort: Desc)])
    @@map("blog_posts")
}

model JobOpening {
    id String @id @default(uuid()) @db.Uuid
    title String
    slug String @unique @db.Citext
    status String
    locationText String? @map("location_text")
    employmentType String @map("employment_type")
    workArrangement String? @map("work_arrangement")
    descriptionRef String @map("description_ref")
    publishedAt DateTime? @map("published_at") @db.Timestamptz(6)
    applications JobApplication[]
    @@index([status,publishedAt(sort: Desc)])
    @@map("job_openings")
}

```

```

model JobApplication {
  id          String @id @default(uuid()) @db.Uuid
  jobId       String @map("job_id") @db.Uuid
  firstName   String @map("first_name")
  lastName    String @map("last_name")
  email       String @db.Citext
  phone       String?
  linkedinUrl String? @map("linkedin_url")
  portfolioUrl String? @map("portfolio_url")
  message     String?
  status      String
  submittedAt DateTime @default(now()) @map("submitted_at") @db.Timestamptz(6)
  job JobOpening @relation(fields:[jobId], references:[id], onDelete:Restrict)
  @@index([jobId,status,submittedAt(sort: Desc)])
  @@map("job_applications")
}

model AuditLog {
  id          String @id @default(uuid()) @db.Uuid
  actorUserId String? @map("actor_user_id") @db.Uuid
  eventType   String @map("event_type")
  targetType  String @map("target_type")
  targetId    String? @map("target_id")
  beforeJson  Json? @map("before_json")
  afterJson   Json? @map("after_json")
  requestId   String? @map("request_id")
  createdAt   DateTime @default(now()) @map("created_at") @db.Timestamptz(6)
  @@index([createdAt(sort: Desc)])
  @@index([actorUserId,createdAt(sort: Desc)])
  @@index([targetType,targetId])
  @@map("audit_logs")
}

```

24. Developer Launch Checklist

Gate	Required before production
Tenant isolation	Automated tests prove cross-workspace access is denied for every protected resource.
RBAC	Permission tests cover user, admin, sub-admin and restricted roles.
Billing	Plan/price/entitlement values render from backend and reconcile with provider IDs.
Credits	Ledger reconciliation test passes; retries cannot double-charge.
GEO	Prompt/run/platform/citation history persists and usage is metered.
Webhooks	Signature verification + unique provider event IDs + async processing.
Uploads	Private storage, MIME/size validation, malware scan for resumes and user files.
Audit	Consequential admin/security/config changes create audit logs.
Backups	Automated PostgreSQL backups + restore test + object-storage lifecycle documented.
Migrations	Staging migration rehearsal passes; rollback/forward-fix plan documented.
Observability	Failed jobs, webhook backlog, database errors and slow queries are observable.
Privacy	Deletion/retention jobs reflect approved policy and do not destroy required financial/audit history.

FINAL
VALIDATION

This design is intentionally production-oriented but should be validated against the exact Better Auth adapter version, billing provider configuration, chosen SEO/GEO data providers and final legal retention policy before the first production migration.